

Roger Frazier, Alex Caramanico, Dereck Roman Rosario, John Kallis

2/8/2023

MATH 341

Homework 1

“=====

Problem 1)

Roger: After I opened Matlab, I typed `doc` on the command window to open the Matlab help browser. Then I clicked on Matlab in the table-of-contents in the left navigation pane to expand its sub-sections. After that, I looked at most of the tutorials and practiced them on Matlab. I did “Desktop Basics, Matrices and Arrays, Array Indexing, Workspace Variables, Text and Characters, calling Functions, 2-D and 3-D Plots, and Help and Documentation”. In the Help and Documentation section, you can see how to obtain documentation information from Matlab. You can open the function documentation in a separate window using the **`doc`** command. You can display function hints in the Command Window by pausing after you type the open parentheses for the function input arguments, example **`mean(`**. Finally you can view an abbreviated text version of the function documentation in the Command Window using the **`help`** command.

Alex: I installed Octave, opened the CLI program, and typed in “`doc`” to bring up the manual. The main thing I took note of was the “Commands for Getting Help” (2.3) section, which will be useful when I run into any problems in coding. Generally, the language seems similar to those I’ve used in the past such as R or Python, but I will have to adjust to using the hyphen key in commands. It seemed worthwhile to familiarize myself with matrix entry and manipulation, so I studied those portions of the manual as well. I also read through the functions section because it will probably be useful later on.

Dereck: I started by renewing my Matlab license which had expired, and then typed “`doc`” in the Matlab command prompt and went to the “MATLAB” section of the table of contents on the left. After taking 426, I was familiar with most of the basic content and jumped around a bit. I took a quick moment to review Matrices and Arrays, as well as Array Indexing under the Get Started section, running some of the examples myself to make sure I was still good with the basics. Since this is a numerical analysis course that extends concepts and ideas taught or addressed in linear algebra, I went to the Linear Algebra section under Mathematics, and I began to familiarize myself with Matrix Operations and Matrix Decomposition commands. After that I looked back at the homework and realized that I missed reading out on Language Fundamentals, and I promptly went back to the aforementioned section and reviewed more basic commands.

John: I googled UMBC MATLAB and found the installation instructions. Then I did the same for Octave because the MatLab license expires. I took MATH 426 in the past so I am familiar with MATLAB. I have also used R, SAS, python, and other programming languages. I googled “Octave Docs” and found the documentation on their website. There is a “Brief overview of Octave” section that I browsed through and looked at the syntax and basic functions. I also went to the “function and scripts” section to look at how to define and call functions in octave. I also looked at how to plot data and entered and ran some values to plot.

“=====

Problem 2)

a) This problem is hand-written.

Some algebra that happens on screen

$$A^{(1)} x^{(1)} = b^{(1)}$$

$$\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 0 & -1 \\ 2 & 2 & 1 & 0 & 1 \\ 2 & -2 & 4 & 2 & 2 \end{bmatrix} \xrightarrow{L=1} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 2 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 0 & 1 \\ 2 & -2 & 4 & 2 & 2 \end{bmatrix} \xrightarrow{L=2} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 2 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 0 & 1 \\ 2 & -6 & 0 & 2 & 4 \end{bmatrix}$$

$$\xrightarrow{L=2} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 2 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \\ 3 & 2 & -2 & -1 & 5 \\ 4 & -6 & 6 & 8 & 10 \end{bmatrix} \xrightarrow{L=3} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 2 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \\ 2 & -2 & -1 & 2 & 5 \\ 4 & -6 & -6 & 20 & 40 \end{bmatrix}$$

$x_4 = 2, x_3 = -1, x_2 = 0, x_1 = 1$

$$v^{(1)} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

Checking:

- $0 - 1 + 2 = 1 \checkmark$
- $1 + 0 - 2 = -1 \checkmark$
- $2 + 0 - 1 = 1 \checkmark$
- $2 + 0 - 1 + 1 = 2 \checkmark$

$$A^{(2)} x^{(2)} = b^{(2)}$$

$$\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 0 & 1 & 1 & 2 \\ 2 & 1 & -1 & 1 & 1 \\ 4 & -1 & -2 & 2 & 0 \\ 3 & -1 & -1 & 2 & -3 \end{bmatrix} \xrightarrow{L=1} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 0 & 1 & 1 & 2 \\ 2 & 1 & -1 & 1 & 1 \\ 4 & -1 & -2 & 2 & 0 \\ 3 & -1 & -1 & 2 & -3 \end{bmatrix} \xrightarrow{L=2} \begin{array}{c} 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 0 & 1 & 1 & 2 \\ 2 & 1 & -1 & 1 & 1 \\ 4 & -1 & -2 & 2 & 0 \\ 3 & -1 & -1 & 2 & -3 \end{bmatrix}$$

$0 \neq -4$, no solution

ellipses & enter
useful for readability

b) In Matlab: For x_1 , we set A_1 and b_1 to the given matrices first then we use $A_1 \backslash b_1$ to get our x_1 . After running the code we get an x_1 that is the same as our hand written work in problem 2a. Then for x_2 , we set A_2 and b_2 to the given matrices first then we use $A_2 \backslash b_2$ to get our x_2 . After running the code we get an x_2 that contains NaN flags and infinite values, and therefore gives no solution which is the same as our hand written work in problem 2a.

Alex: Completed problem using Octave, where x_2 was calculated to "machine accuracy", compared to Matlab's "singular to working precision."

This is the m-file:

```
format compact
diary hw1_prob_2a.dia
```

```
A_1 = [0 1 1 1; ...
       1 2 2 0; ...
       2 2 1 0; ...
       2 -2 4 2]
```

```
b_1 = [1; -1; 1; 2]
```

```
x_1 = A_1\b_1
```

```
A_2 = [1 1 0 1; ...  
       2 1 -1 1; ...  
       4 -1 -2 2; ...  
       3 -1 -1 2]
```

```
b_2 = [2; 1; 0; -3]
```

```
x_2 = A_2\b_2
```

```
diary off
```

This is the diary file:

```
A_1 =  
    0     1     1     1  
    1     2     2     0  
    2     2     1     0  
    2    -2     4     2
```

```
b_1 =  
    1  
   -1  
    1  
    2
```

```
x_1 =  
    1  
    0  
   -1  
    2
```

```
A_2 =  
    1     1     0     1  
    2     1    -1     1  
    4    -1    -2     2  
    3    -1    -1     2
```

```
b_2 =  
    2  
    1  
    0  
   -3
```

```
[Warning: Matrix is singular to working precision.]
```

```
[> In <a
```

```
href="matlab:matlab.internal.language.introspective.errorDocCallback('hw1  
_prob_2a', 'C:\Users\roger\OneDrive\Documents\MATLAB\MATH  
341\Hw1\hw1_prob_2a.m', 19)" style="font-weight:bold">hw1_prob_2a</a> (<a  
href="matlab: opentoline('C:\Users\roger\OneDrive\Documents\MATLAB\MATH  
341\Hw1\hw1_prob_2a.m',19,0)">line 19</a>)
```

```
]
```

x_2 =
NaN
NaN
Inf
-Inf

c) From the Department of Economics (Western Social Science): A unique solution to a system of linear equations exists if and only if:

- i) The number of unknowns equals the number of equations
- ii) All equations are consistent (e.g., $2x_1 + x_2 = 3$ is not consistent with $4x_1 + 2x_2 = 5$)
- iii) There is no linear dependence between any two or more of the equations (i.e. they are all independent of each other)

If condition iii holds, the matrix is non-singular, and potentially invertible. This brings about the Invertible Matrix Theorem, which states the following according to Wolfram MathWorld:

The $n \times n$ matrix $\mathbf{A}$ is invertible if and only if any (and hence all) of the following hold:

1. $\mathbf{A}$ is row-equivalent to the $n \times n$ identity matrix
2. $\mathbf{A}$ has n pivot positions
3. The equation $\mathbf{Ax} = \mathbf{0}$ has only trivial solution $\mathbf{x} = \mathbf{0}$
4. The columns of $\mathbf{A}$ form a linearly independent set
5. The linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$ is one-to-one
6. For each column vector $\mathbf{b}$, the equation $\mathbf{Ax} = \mathbf{b}$ has a unique solution
7. The columns of $\mathbf{A}$ span R^n
8. The linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$ is a surjection
9. There is an $n \times n$ matrix $\mathbf{C}$ such that $\mathbf{C}^*\mathbf{A} = I_n$
10. There is an $n \times n$ matrix $\mathbf{D}$ such that $\mathbf{A}^*\mathbf{D} = I_n$
11. The transpose matrix $\mathbf{A}^T$ is invertible
12. The columns of $\mathbf{A}$ form a basis R^n
13. The column space of $\mathbf{A}$ is equal to R^n
14. The dimension of the column space of $\mathbf{A}$ is n
15. The rank of $\mathbf{A}$ is n
16. The null space of $\mathbf{A}$ is $\{0\}$
17. The dimension of the null space of $\mathbf{A}$ is 0
18. 0 fails to be an eigenvalue of $\mathbf{A}$
19. The determinant of $\mathbf{A}$ is not zero
20. The orthogonal complement of the column space of $\mathbf{A}$ is $\{0\}$
21. The orthogonal complement of the null space of $\mathbf{A}$ is R^n
22. The row space of $\mathbf{A}$ is R^n
23. The matrix $\mathbf{A}$ has n non-zero singular values